USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

1) Please fill out the Engineering Change Request Form on the following pages completely:
   a) Detail the names and contact details for each of the ECR contributors
   b) Update the ECR Title
   c) Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
2) For each section/table/figure to be updated:
   a) Detail the section number, starting page and figure/table number to be updated as appropriate.
   b) Detail existing text under "From Text"
   c) Detail changed text under "To Text"
3) Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step 1)b)
4) Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
   a) This ECR will then be reviewed by the Power Delivery Compliance Working Group.
   b) Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: VPD test update – TEST.PD.PROT.ALL3.X Tests
Applied to: USB PD CTS Q3, 2026

Brief description of the functional changes proposed:

VPD responds to SOP' messages only. It shall not respond to other USB PD Messaging per spec definition(See below), therefore SOP extended message checks in TEST.PD.PROT.ALL.5, TEST.PD.PROT.ALL.3.1-3.7 not applicable for VPD DUTs and need to be skipped for VPD DUTs

---------Spec Reference----------

Type-C Spec section 4.10.2:

"A VCONN-Powered USB Device shall respond to USB PD messaging on SOP' and shall not respond to other USB PD messaging."

Benefits as a result of the proposed changes:

VPDs correctly tested in alignment with base spec definition.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:

None

An analysis of the hardware implications:

None

An analysis of the software implications:

None

An analysis of the compliance testing implications:

Tester to update the corresponding tests per the ECR updates.

An analysis of the Vendor Info File (VIF) implications:

None

USB PD CTS ENGINEERING CHANGE NOTICE

Actual Change Requested

Update Test Conditions table in the sections below.

To Text:

5.1.1 PD2 and PD3 Modes

The tests in this subsection are run firstly in PD2 mode and secondly in PD3 mode.
The tests in this subsection are applicable to UUTs listed in the Test Conditions tables of the respective test..

TEST.PD.PROT.ALL.5 Unrecognized Message

1. There are 9 possible bring-up procedures:
    a. The UUT has VIF field PD_Port_Type set to Consumer Only. The Tester behaves as a Source only and it runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2
    b. The UUT has VIF field PD_Port_Type set to Consumer/Provider, each test under this section needs to be run twice, first time the Tester behaves as a Source only and it runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2; second time the Tester behaves as a Provider/Consumer and it runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.7 (Dual-Role Power bit B29 in the single PDO is set to 1b).
    c. The UUT has VIF field PD_Port_Type set to Provider Only. The Tester behaves as a Sink only and it runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.
    d. The UUT has VIF field PD_Port_Type set to Provider/Consumer, each test under this section needs to be run twice, first time the Tester behaves as a Consumer/Provider and it runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.8; second time the Tester behaves as a Sink Only and it runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1 (Dual-Role Power bit B29 in the single PDO is set to 1b).
    e. The UUT has VIF field PD_Port_Type set to DRP, each test under this section needs to be run twice, first time the Tester behaves as a Source only and it runs bring-up procedure with the UUT as a Sink COMMON.PROC.BU.2; second time the Tester behaves as a Sink only and it runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1 (Dual-Role Power bit B29 in the single PDO is set to 0b).
    f. The UUT has VIF field VIF_Product_Type set to Cable. The Tester runs bring-up procedure with the UUT as a Cable Plug COMMON.PROC.BU.3
    ~~g. The UUT has VIF field VIF_Product_Type set to Port Product and PD_Port_Type set to eMarker. The Tester runs bring-up procedure with the UUT as a VPD COMMON.PROC.BU.10~~
    h. The UUT has VIF field PD_Port_Type set to DRP, Consumer/Provider or Consumer only, and VIF parameter Captive_Cable is set to YES, and VIF parameter eMarked Captive Cable is set to YES*. The Tester runs bring-up procedure with the UUT as a Sink UUT for Captive Cable Test COMMON.PROC.BU.6.
    i. The UUT has VIF field PD_Port_Type set to Provider/Consumer or Provider only, and VIF parameter Captive_Cable is set to YES, and VIF parameter eMarked Captive Cable is set to YES*. The Tester runs bring-up procedure with the UUT as a Source UUT for Captive Cable Test COMMON.PROC.BU.11.

5.1.2 PD3 Mode

The tests in this subsection are run only in PD3 mode.
The tests in this subsection are applicable to UUTs listed in the Test Conditions tables of the respective test.

TEST.PD.PROT.ALL3.1 Get_Status Response

TEST.PD.PROT.ALL3.2 Get_Manufacturer_Info Response

TEST.PD.PROT.ALL3.3 Invalid Manufacturer Info Target

TEST.PD.PROT.ALL3.4 Invalid Manufacturer Info Ref

TEST.PD.PROT.ALL3.5 Chunked Extended Message Response

TEST.PD.PROT.ALL3.6 ChunkSenderResponseTimer Timeout

TEST.PD.PROT.ALL3.7 Security Messages Supported

Remove the following bring-up procedure from Step1 of all of the tests listed above.

~~The UUT has VIF field VIF_Product_Type set to Port Product and PD_Port_Type set to eMarker. The Tester runs bring-up procedure with the UUT as a VPD COMMON.PROC.BU.10~~